IEAP-Series03-Rstudio: Data wrangling and ANOVA

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GitHub repository: https://github.com/ElouanDef/IEAP-Series03-RStudio.git

Team workflow with Git

We worked as a team of three using GitHub to organize and share our work. Each member worked on a separate branch and used commits and Pull Requests to merge their work into the main branch.

For Shunan and Elouan, it was the first time working on a GitHub project organized into several sub-documents, with one file for each section. We found this method very useful because it made the documents shorter, clearer and easier to organize.

At the end, we combined all the sub-documents into one main Quarto file to create the final report.

Challenges and lessons learned

One of our main challenges was coordinating our work. We all had different schedules, which made it difficult to find time to work together. As a result, completing the assignment took longer than expected.

We also encountered some technical difficulties, particularly with file paths and dependencies between R scripts. Some sections needed objects created in previous sections, which sometimes caused errors when running the code separately.

However, we learned that dividing the project into smaller files was an efficient way to organize our work. We also improved our understanding of GitHub, especially how to use branches, commits and Pull Requests to collaborate.

We did not fully respect the rule of one commit per question: questions 1.4 and 1.5 were committed together. Next time, we will make one commit for each question.

Overall, this project taught us the importance of communication, organization and planning when working as a team.

1 ANOVA: Analysis of variance

1.1 Think before coding

Scientific question: is the effect of difficulty (ID) on movement time (MT) the same in all groups?

1. Visualization: mean MT as a function of ID, one line per GROUP (interaction plot).
2. Test: GROUP × ID interaction in a mixed ANOVA (GROUP between subjects, ID within subjects).
3. Result sentence: see section 1.3.

1.2 Library used

We use `ez` (function `ezANOVA`) for the ANOVA.

```
library(tidyverse)
library(ez)
```

1.3 Data reorganization

Reading data

Each row of the file is one reaching movement.

```
results <- read.table("data/Results.txt", header = TRUE,
                      sep = " ", stringsAsFactors = TRUE)
names(results)
```

```
[1] "GROUP"        "SUBJ"         "TRIAL"        "ORI"          "DIR"
[6] "ID"           "REP"          "NbVelPeaks"   "MovementTime"
```

Number of measures per GROUP and ID

```
results %>% count(GROUP, ID)
```

```
  GROUP  ID    n
1    ap ID1  740
2    ap ID2  740
3    ap ID3  740
4    cp ID1 1140
5    cp ID2 1140
6    cp ID3 1140
7    pp ID1  760
8    pp ID2  760
9    pp ID3  400
```

Data are missing for ID3 in the pp group, so ID3 is removed.

Number of measures per REP

```
results %>% filter(ID != "ID3") %>% count(REP)
```

```
  REP    n
1  R1 2040
2  R2 2040
3  R3 1200
```

```
results %>% filter(ID != "ID3") %>% count(GROUP, REP)
```

```
  GROUP REP   n
1    ap  R1 520
2    ap  R2 520
3    ap  R3 440
4    cp  R1 760
5    cp  R2 760
6    cp  R3 760
7    pp  R1 760
8    pp  R2 760
```

R3 is missing in the pp and ap groups, so R3 is removed.

Clean data

```
results_clean <- results %>%
  filter(ID != "ID3", REP != "R3") %>%
  droplevels()

# Must be balanced
results_clean %>% count(GROUP, ID, REP)
```

```
   GROUP  ID REP   n
1     ap ID1  R1 260
2     ap ID1  R2 260
3     ap ID2  R1 260
4     ap ID2  R2 260
5     cp ID1  R1 380
6     cp ID1  R2 380
7     cp ID2  R1 380
8     cp ID2  R2 380
9     pp ID1  R1 380
10    pp ID1  R2 380
11    pp ID2  R1 380
12    pp ID2  R2 380
```

Number of participants per group

```
results_clean %>%
  distinct(GROUP, SUBJ) %>%
  count(GROUP)
```

```
  GROUP  n
1    ap 13
2    cp 19
3    pp 19
```

One value per participant and ID

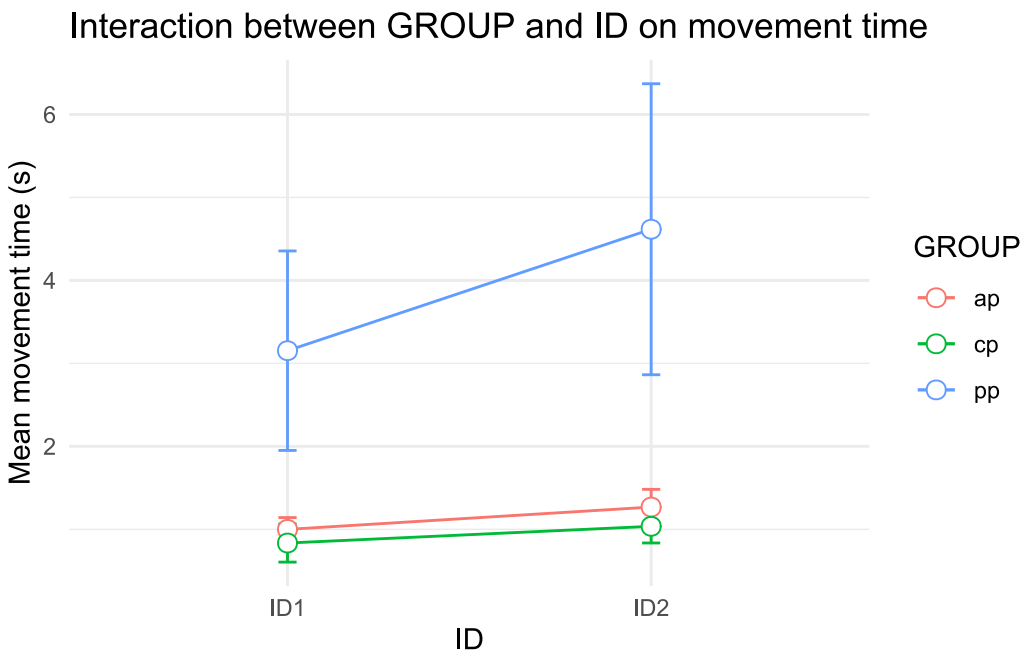
The ANOVA needs a single value per participant and per condition, so we average the movements.

```
data_anova <- results_clean %>%
  group_by(GROUP, SUBJ, ID) %>%
  summarise(MT = mean(MovementTime), .groups = "drop") %>%
  mutate(SUBJ = factor(SUBJ))
```

Visualization of the interaction

```
# Mean ± 0.5 SD, as in the assignment (mean_sdl with mult = 0.5)
mean_sd_half <- function(y) {
  data.frame(y = mean(y), ymin = mean(y) - sd(y) / 2, ymax = mean(y) + sd(y) / 2)
}

ggplot(results_clean, aes(x = ID, y = MovementTime, color = GROUP)) +
  stat_summary(fun = mean, geom = "line", aes(group = GROUP)) +
  stat_summary(fun.data = mean_sd_half, geom = "errorbar", width = 0.05) +
  stat_summary(fun = mean, geom = "point", shape = 21, fill = "white", size =
3) +
  labs(title = "Interaction between GROUP and ID on movement time",
       x = "ID", y = "Mean movement time (s)") +
  theme_minimal()
```



The lines are not parallel: movement time increases much more with ID in the pp group than in the two other groups.

ANOVA

```
anova_res <- ezANOVA(data = data_anova,
                     dv = MT,
                     wid = SUBJ,
                     within = ID,
                     between = GROUP,
                     detailed = FALSE)
```

```
Warning: Data is unbalanced (unequal N per group). Make sure you specified a
well-considered value for the type argument to ezANOVA().
```

```
anova_res$ANOVA
```

```
    Effect DFn DFd        F            p p<.05        ges
2    GROUP   2  48 37.24408 1.720007e-10     * 0.57462110
3       ID   1  48 30.77618 1.221240e-06     * 0.07667596
4 GROUP:ID   2  48 11.58124 7.866206e-05     * 0.05882293
```

The effect of difficulty depends on the group: the GROUP × ID interaction is significant, F(2, 48) = 11.58, p < .001, generalized eta squared = 0.06. The main effects of GROUP, F(2, 48) = 37.24, p < .001, and of ID, F(1, 48) = 30.78, p < .001, are also significant.

1.4 Run the linear regression by GROUP

```
library(dplyr)

# 1. Clean the data: remove ID3 and keep ID1 and ID2
data_clean <- results_clean %>%
  filter(ID %in% c("ID1", "ID2")) %>%
  mutate(
    ID_numeric = case_when(
      ID == "ID1" ~ 3.8,
      ID == "ID2" ~ 5.43
    )
  ) %>%
  droplevels()

# 2. Verify the ID mapping
print(
  data_clean %>%
    distinct(ID, ID_numeric) %>%
    arrange(ID_numeric)
)
```

```
   ID ID_numeric
1 ID1       3.80
2 ID2       5.43
```

```
# 3. Fit one linear regression for each GROUP
models <- data_clean %>%
  group_by(GROUP) %>%
  group_modify(~ {
    fit <- lm(MovementTime ~ ID_numeric, data = .x)

    tibble(
      intercept = unname(coef(fit)[1]),
      slope = unname(coef(fit)[2]),
      equation = sprintf(
        "MT = %.3f + %.3f * ID",
        unname(coef(fit)[1]),
        unname(coef(fit)[2])
```

```
      )
    )
  }) %>%
  ungroup()

# 4. Display the three equations
print(models)
```

```
# A tibble: 3 × 4
  GROUP intercept slope equation
  <fct>     <dbl> <dbl> <chr>
1 ap        0.372 0.165 MT = 0.372 + 0.165 * ID
2 cp        0.366 0.123 MT = 0.366 + 0.123 * ID
3 pp       -0.259 0.898 MT = -0.259 + 0.898 * ID
```

```
# 5. Export the equations to the repository
write.csv(
  models,
  file = "Regression_Equations.csv",
  row.names = FALSE
)

writeLines(
  paste(models$GROUP, models$equation, sep = ": "),
  con = "Regression_Equations.txt"
)
```

1.5 Graph for publication

```
library(dplyr)
library(ggplot2)

# 1. Calculate group means and standard deviations
summary_data <- data_clean %>%
  group_by(GROUP, ID_numeric) %>%
  summarise(
    MeanMT = mean(MovementTime, na.rm = TRUE),
    SD = sd(MovementTime, na.rm = TRUE),
    .groups = "drop"
  )

# 2. Fit a separate regression for each group
```

```r
models <- data_clean %>%
  group_by(GROUP) %>%
  group_modify(~ {
    fit <- lm(MovementTime ~ ID_numeric, data = .x)

    tibble(
      intercept = unname(coef(fit)[1]),
      slope = unname(coef(fit)[2])
    )
  }) %>%
  ungroup()

# 3. Generate regression lines across the graph
prediction_data <- models %>%
  group_by(GROUP) %>%
  group_modify(~ {
    x <- seq(-0.3, 6.3, length.out = 200)

    tibble(
      ID_numeric = x,
      PredictedMT = .x$intercept[1] +
        .x$slope[1] * x
    )
  }) %>%
  ungroup()

# 4. Prepare clearly separated equation labels
equations <- models %>%
  mutate(
    label = sprintf(
      "MT = %.3f + %.3f * ID",
      intercept, slope
    ),
    x = 0.5,
    y = case_when(
      GROUP == "pp" ~ 5.45,
      GROUP == "ap" ~ 5.10,
      GROUP == "cp" ~ 4.75
    )
  )

# 5. Match the reference colors
group_colors <- c(
```

```
  "pp" = "red",
  "ap" = "green3",
  "cp" = "blue"
)

# 6. Create the publication-style graph
p <- ggplot() +

  # Dashed regression lines
  geom_line(
    data = prediction_data,
    aes(
      x = ID_numeric,
      y = PredictedMT,
      color = GROUP,
      group = GROUP
    ),
    linetype = "dashed",
    linewidth = 0.55
  ) +

  # Mean +/- standard deviation
  geom_errorbar(
    data = summary_data,
    aes(
      x = ID_numeric,
      ymin = MeanMT - SD,
      ymax = MeanMT + SD,
      color = GROUP
    ),
    width = 0.08,
    linewidth = 0.55
  ) +

  # Mean points
  geom_point(
    data = summary_data,
    aes(
      x = ID_numeric,
      y = MeanMT,
      color = GROUP
    ),
    shape = 1,
```

```
  size = 3
) +

# Solid lines joining observed means
geom_line(
  data = summary_data,
  aes(
    x = ID_numeric,
    y = MeanMT,
    color = GROUP,
    group = GROUP
  ),
  linewidth = 0.7
) +

# Non-overlapping equations
geom_text(
  data = equations,
  aes(
    x = x,
    y = y,
    label = label,
    color = GROUP
  ),
  inherit.aes = FALSE,
  hjust = 0,
  vjust = 0.5,
  size = 3.6
) +

scale_color_manual(
  values = group_colors,
  breaks = c("pp", "ap", "cp"),
  name = "GROUP"
) +

# Numeric ID axis
scale_x_continuous(
  breaks = 0:6,
  expand = expansion(mult = 0),
  limits = c(-0.3, 6.3)
) +
```

```
  # Movement time in seconds
  scale_y_continuous(
    breaks = -1:6,
    expand = expansion(mult = 0)
  ) +

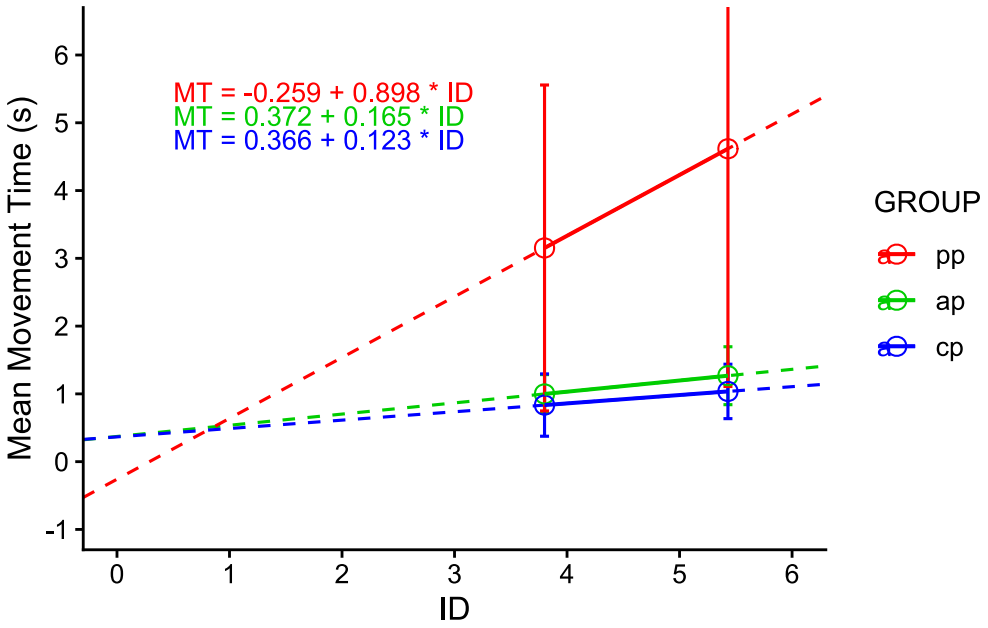
  coord_cartesian(
    ylim = c(-1.3, 6.7)
  ) +

  labs(
    x = "ID",
    y = "Mean Movement Time (s)"
  ) +

  theme_classic(base_size = 12) +

  theme(
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10, color = "black"),
    axis.line = element_line(linewidth = 0.5),
    axis.ticks = element_line(linewidth = 0.5),
    legend.position = "right",
    legend.title = element_text(size = 11),
    legend.text = element_text(size = 10),
    plot.margin = margin(10, 15, 10, 10)
  )

# Display
print(p)
```



1.6 Export the graph as a PDF

```
dir.create("figures", showWarnings = FALSE)

# Save the regression plot as a PDF
ggsave(
  filename = "figures/MovementTime_Regression.pdf",
  plot = p,
  width = 8,
  height = 6,
  units = "in",
  device = "pdf"
)
```

1.7 Discuss the differences with the article

Our results are slightly different from those presented in the scientific article. The regression equations and the ANOVA values are not exactly the same because we used a simpler statistical method and excluded some incomplete experimental conditions (ID3 and REP3).

However, the main conclusions are similar.

In our graph, movement time increases with the Index of Difficulty (ID) in all three groups. However, the increase is much stronger in the patient group (slope = 0.898) than in the age-matched group (slope = 0.165) and the young group (slope = 0.123).

This result is consistent with Figure 3 of the article, where the patient group also shows a much steeper regression slope than the two healthy groups.

Furthermore, our ANOVA shows a significant Group × ID interaction ($F(2,48) = 11.58$, $p < 0.001$). This confirms that the effect of task difficulty on movement time differs between groups.

Therefore, although the numerical results are different, our simpler analysis leads to the same main scientific conclusion : Increasing task difficulty has a much stronger effect on movement time in patients with stroke than in healthy participants.

Data sources

- `Results.txt`: data provided on the course Moodle page, https://moodle.umontpellier.fr/mod/folder/view.php?id=1025707

References

Mottet, D., van Dokkum, L. E. H., Froger, J., Gouaïch, A., & Laffont, I. (2017). Trajectory formation principles are the same after mild or moderate stroke. *PLOS ONE*, 12(3), e0173674. https://doi.org/10.1371/journal.pone.0173674

Checklist

- ☒ Public repository named `IEAP-Series03-RStudio`, `main` branch up to date
- ☒ `README.md` clear and complete
- ☒ `LICENSE` file
- ☒ Master document `IEAP-Series03-Rstudio.qmd` and its PDF
- ☒ Authors and date in the header
- ☒ Clickable link to the public repository
- ☒ Analysis of the team workflow with Git
- ☒ Challenges and lessons learned
- ☒ Complete code and answers to questions 1.1 to 1.7
- ☒ Data sources with URLs
- ☒ Scientific references with DOIs
- [] At least one commit per question, with meaningful messages
- ☒ One branch per task, merged into `main` through pull requests
- ☒ No TODO left in the document
- ☒ PDF uploaded on Moodle